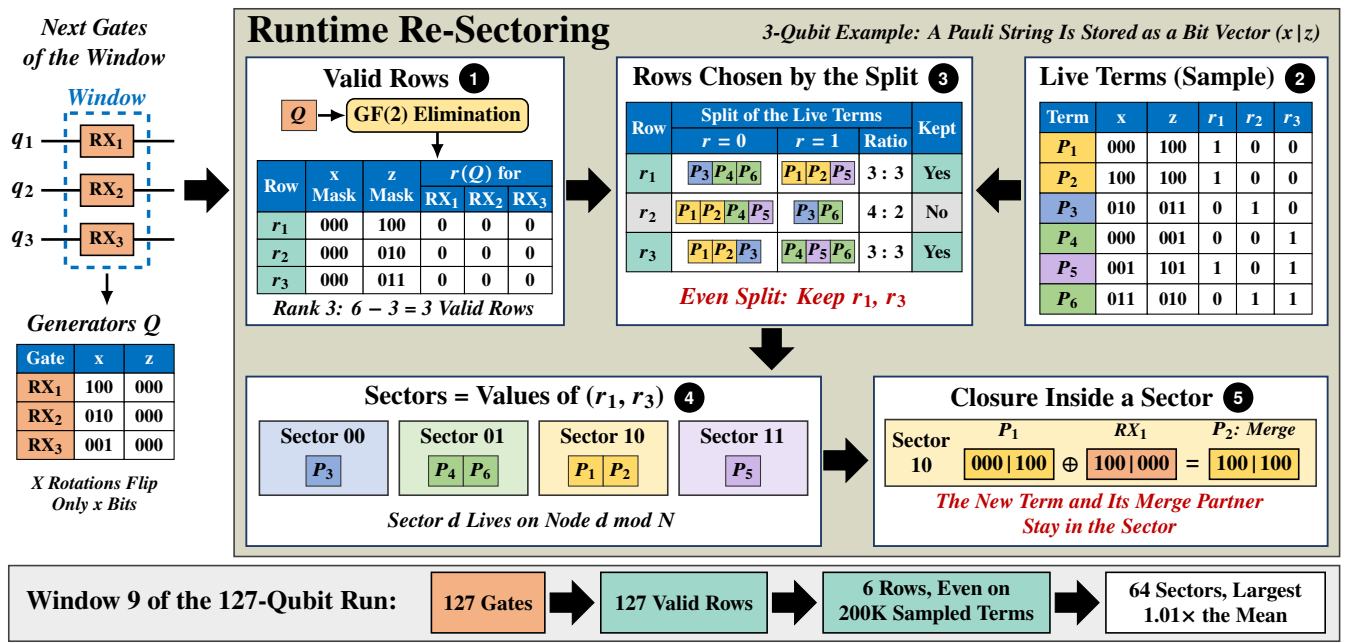

Next Gates of the Window
Window
q_1
q_2
q_3
RX1
RX2
RX3
Generators Q
Gate | x | z
RX1 | 100 | 000
RX2 | 010 | 000
RX3 | 001 | 000
X Rotations Flip Only x Bits
Runtime Re-Sectoring
3-Qubit Example: A Pauli String Is Stored as a Bit Vector (x | z)
Valid Rows 1
Q → GF(2) Elimination
Row | x Mask | z Mask | r(Q) for RX1 | RX2 | RX3
r1 | 000 | 100 | 0 | 0 | 0
r2 | 000 | 010 | 0 | 0 | 0
r3 | 000 | 011 | 0 | 0 | 0
Rank 3: 6 − 3 = 3 Valid Rows
Rows Chosen by the Split 3
Row | Split of the Live Terms r = 0 | r = 1 | Ratio | Kept
r1 | P3 P4 P6 | P1 P2 P5 | 3 : 3 | Yes
r2 | P1 P2 P4 P5 | P3 P6 | 4 : 2 | No
r3 | P1 P2 P3 | P4 P5 P6 | 3 : 3 | Yes
Even Split: Keep r1, r3
Live Terms (Sample) 2
Term | x | z | r1 | r2 | r3
P1 | 000 | 100 | 1 | 0 | 0
P2 | 100 | 100 | 1 | 0 | 0
P3 | 010 | 011 | 0 | 1 | 0
P4 | 000 | 001 | 0 | 0 | 1
P5 | 001 | 101 | 1 | 0 | 1
P6 | 011 | 010 | 0 | 1 | 1
Sectors = Values of (r1, r3) 4
Sector 00 P3
Sector 01 P4 P6
Sector 10 P1 P2
Sector 11 P5
Sector d Lives on Node d mod N
Closure Inside a Sector 5
Sector 10
P1 000 | 100 ⊕ RX1 100 | 000 = P2: Merge 100 | 100
The New Term and Its Merge Partner Stay in the Sector
Window 9 of the 127-Qubit Run:
127 Gates
127 Valid Rows
6 Rows, Even on 200K Sampled Terms
64 Sectors, Largest 1.01× the Mean